

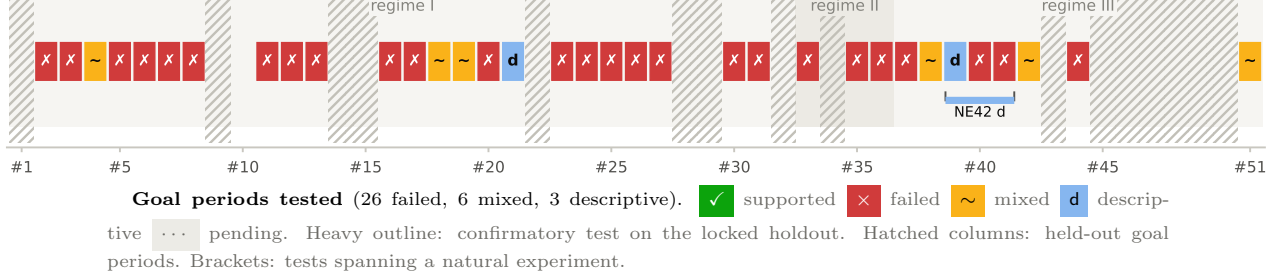
H114: A Griffiths phase: rare strong pairs give a subcritical swarm heavy-tailed talk

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

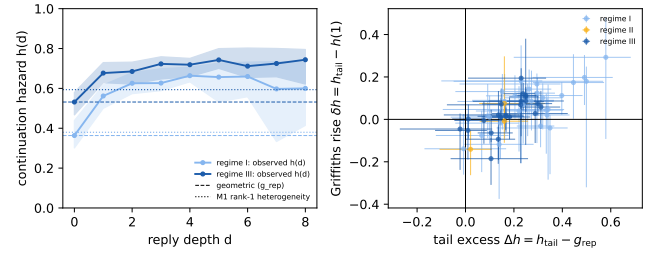
Bottom line. Reply chains are heavier than uniform subcritical branching because of first-order thread momentum, not because rare strong agent pairs form locally supercritical regions.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH346 proposed a Griffiths phase: a swarm subcritical on average whose rare strongly coupled pairs make reply chains heavy-tailed. H114 measures reply depth on DQ2 parent links in 66 non-holdout units (54 usable). Chains are heavier than a uniform subcritical process: the tail continuation hazard exceeds the reply branching ratio in 52/54 units (CI > 0 in 33/54). The excess sits at depth 1: a reply's parent is itself a reply more often than the mean allows (+0.16, 54/54 units). The hazard then rises little (significant in 10/54). Strong pairs (pair gain > 0.5) exist in 18/66 units, are half of them readers answering three hub authors, and carry the tail in 1/16. **The Griffiths mechanism fails**; thread momentum explains the tail. The holdout is not run.



Observables. Left: continuation hazard $h(d)$ by reply depth (median and IQR over units), against the geometric law at g_{rep} (dashed) and the rank-1 agent model M1 (dotted). The hazard jumps at depth 1 and stays nearly flat. Right: per-unit tail excess Δh against the Griffiths rise δh . Most units have $\Delta h > 0$ with δh near 0.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

A reply forest (models 09, 03, 01). Each agent message has at most one agent parent. Its depth D counts links to the root. A uniform subcritical process gives a geometric depth law at the reply branching ratio g_{rep} . A Griffiths phase mixes continuation rates: strong pairs (i, j) with pair gain g_{ij} near 1 make the hazard $h(d)$ rise with depth. Thread momentum makes h jump at $d = 1$ and stay flat. M1 and M2 are first-order author chains without and with pair-specific transitions.

$$S(d) = P(D \geq d), \quad h(d) = \frac{S(d+1)}{S(d)}, \quad g_{\text{rep}} = S(1)$$

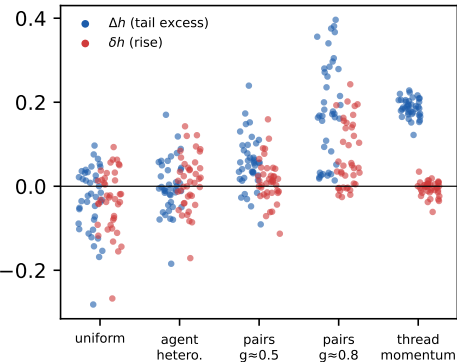
$$h_{\text{tail}} = \frac{\sum_{d \geq 4} S(d+1)}{\sum_{d \geq 4} S(d)}, \quad \Delta h = h_{\text{tail}} - g_{\text{rep}}, \quad \delta h = h_{\text{tail}} - h(1)$$

$$g_{ij} = L_{ij}/R_{ij}, \quad \text{Mk} : s_d = p \odot (P_k s_{d-1})$$

H114-griffiths-phase-pairs · generated 2026-10-04 15:03 by infra/summaries/build_summaries.py · card hypotheses/H114-griffiths-phase-pairs/README.md · ratings: H114 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 $\Delta h > 0$ in $\geq 2/3$	CI 33/54; point 52/54	failed narrowly
P2 rise $\delta h > 0$ in $\geq 1/2$	10/54 (median +0.05)	failed
P3 strong pair in $\geq 1/2$ of units	18/66; 37 pairs; 4 mutual	failed
P4 strong cut carries the tail	1/16	failed
P5 M2 beats M1	39/54, M2-M1 +0.003	null in size
P6 tail above M2 (momentum)	31/54	supported
P7 $\rho(\Delta h, \text{strong share}) \geq 0.4$	-0.14; exponent unidentifiable	failed
P8 naming $\geq 1.2\times$; no family	0.73 $\times$; OR 0.31	mixed
G51 assigned pairs strong / higher g	0/176 / MW p 0.0003	failed / yes



Second figure. Synthetic forests planted on real message skeletons (4 units, 10 replicates). Uniform and agent-heterogeneity worlds give $\Delta h \leq 0$. Strong pairs at $g \approx 0.8$ give $\Delta h > 0$ and a rise $\delta h > 0$; at $g \approx 0.5$ they are invisible. Thread momentum gives $\Delta h \approx 0.19$ with no rise.

GOAL PERIODS IN DETAIL

G02 $\times$ failed
G03 $\times$ failed
G04 $\sim$ mixed
G05 $\times$ failed
G06 $\times$ failed
G07 $\times$ failed
G08 $\times$ failed
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
+25 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

CAVEATS

DQ2’s parent rule favors named partners and its reply probability keeps a thread-membership component, so label-driven momentum is possible (R4); a label-free reply proxy is needed. Pairs with $g \approx 0.5$ are not detectable at these counts (Amendment A1). One parent per message projects a richer reply graph. P1’s CI count (61%) misses its 2/3 bar. Predictions were written knowing H62’s thread-field result.

ROUND 2

Rebuild chains from ledger timing without DQ2 labels; split momentum into read vs in-flight parents; model hub authors as fields.

- **Essence.** whether swarm talk has a disordered, locally near-critical structure an operator could target, or only conversation-level memory.
- **H114-R1** Label-free momentum.
- **H114-R2** Momentum as a field or a coupling.
- **H114-R3** Hubs.
- **H114-R4** Holdout.